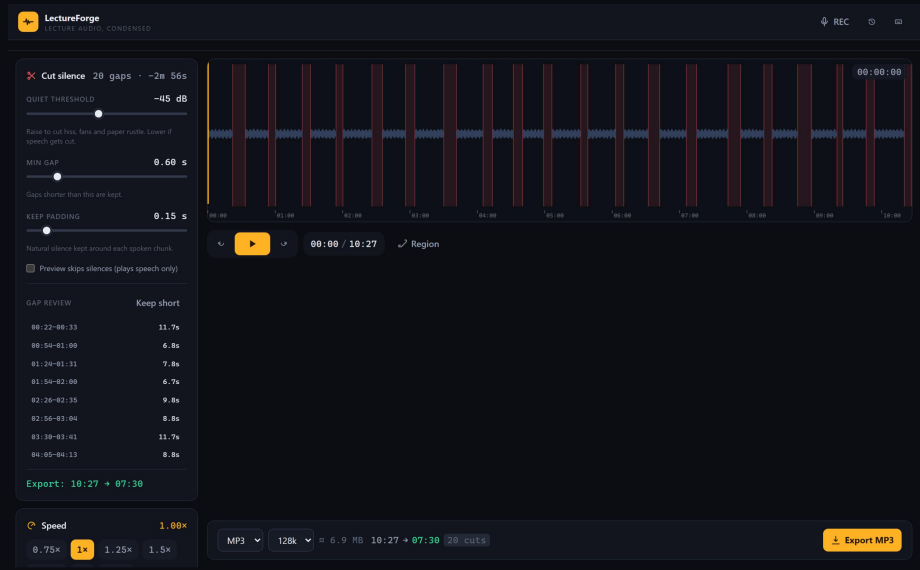
LectureForge

Cut the silence. Keep the lecture.

A browser app that removes dead air from class recordings, speeds them up without a chipmunk voice, and exports MP3 —
with no upload, no account, and no installation.



20 silent gaps found automatically in a 10-minute lecture

Nash Bruce V. Quiros · BSCS-3 · Multimedia · November 4, 2026

One hour of lecture. Up to 15 minutes of dead air.

Silence while a class settles, the teacher thinks, or papers shuffle. Hand-cutting it in an editor means finding every silent part yourself.

10–15 min

of silence in a typical 1-hour class recording

20+ edits

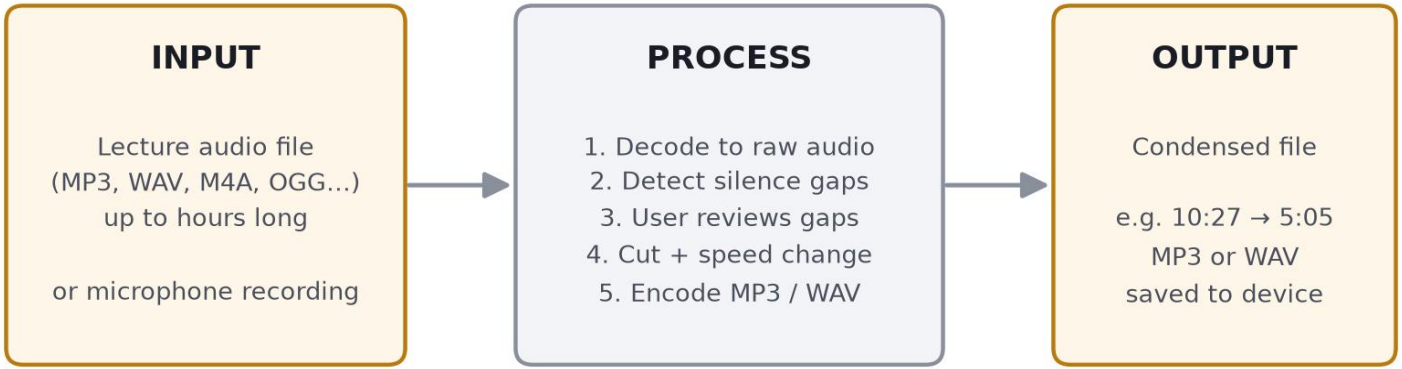
to select and delete each gap by hand

15+ min

of careful zooming, selecting, deleting, exporting

Who it helps: students reviewing recorded lectures, and teachers sharing class recordings on slow connections.

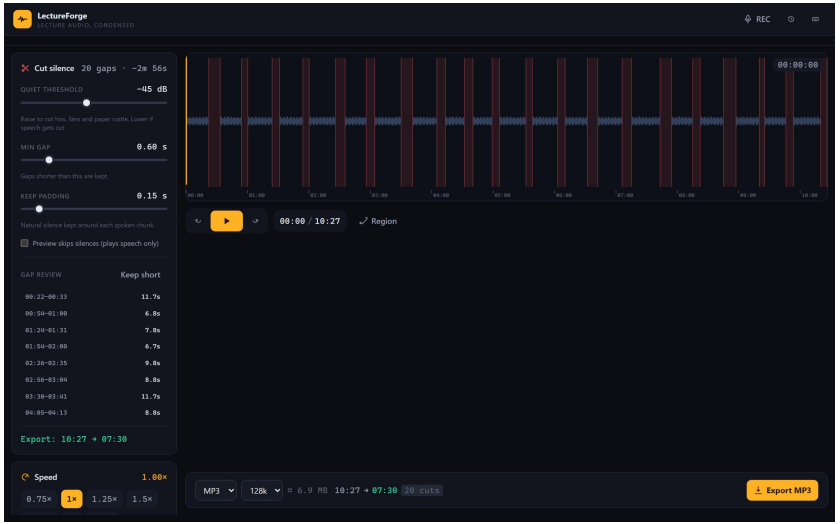
How it works



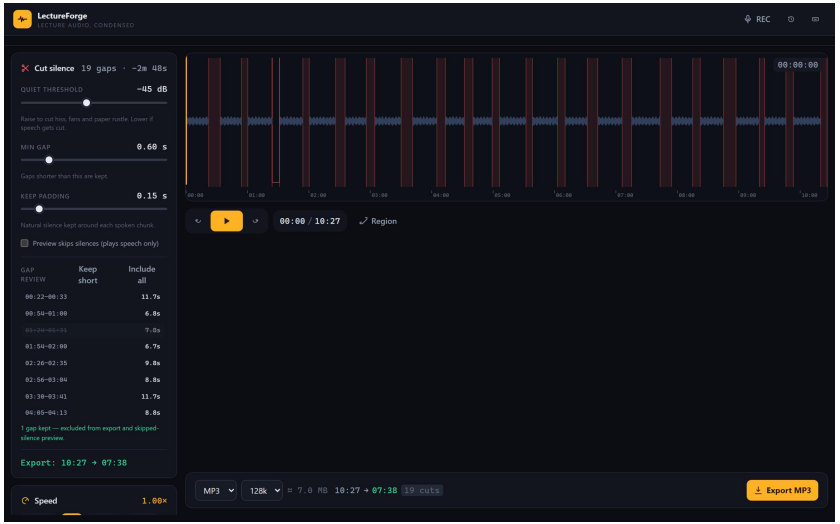
All steps run locally in the browser (Web Workers) — the audio never leaves the device.

Loudness is measured in small frames; silent gaps become visible cuts the user reviews before export.

Every cut visible — every cut reviewable



Gaps are marked red on the waveform — found in 2.3 seconds



One click keeps a pause; the estimate updates instantly

Plus: batch queue, microphone recording, offline install (PWA), keyboard shortcuts.

Tested end to end

All five manual tests passed on a 10.4-minute sample lecture (105 MB, 20 inserted silences):

2.3 s	20 / 20	5:05	25 / 25
to load and analyze the file	silent gaps detected	exact export length	automated engine tests pass

- Keeping one gap in review updated the plan from 20 cuts to 19 (10:27 → 07:38) — and the export honored it.
- MP3 at 1.5× speed finished in 20.5 s: valid MP3, duration exactly 5:05, natural pitch.
- Accessibility audit (axe-core): zero serious findings across all app states; loudness leveling verified.

Same task, two ways

Cut the silence from the same 10.4-minute lecture

LectureForge

3 clicks · ~23 seconds

- Drop the file (2.3 s analysis)
- Review gaps — optional
- Export MP3 (20.5 s)

Audacity (manual way)

60+ interactions · 15+ minutes

- Zoom in, hunt each silence by eye and ear
- Select + delete — 20 times over
- One mistake can clip a word

Measured: LectureForge timings are from the tests; the Audacity path is the documented manual workflow, estimated.

Conclusion



What works

- All 5 manual tests passed
- 20-min lecture → 5:05 in ~23 s
- Loudness leveling + mobile layout
- Fully offline & private

Limits

- Audio only — no video files
- Recorder needs mic permission
- Files >220 MB at lower rate
- Safari 15+ only

Next

- Video soundtrack support
- Chapter markers in exports
- Filipino interface
- iOS Safari polish